SIMATS ENGINEERING

CONTINOUS INTERNAL ASSESSMENT- SHORT ANSWER TEXT

COURSE NAME: DATABASE MANAGEMENT SYSTEMS
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)

Questions:05 **Total marks:100**

Annexure A

SHORT ANSWER TEXT

Code of Conduct:

I R Charudatha / 192260033 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

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DATABASE MANAGEMENT SYSTEMS

CO1AT1

Short Answers

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1)

1. Supertype and subtypes

   * Supertype : Vehicle
   * Subtypes : Car and Truck
   * The Vehicle is specialized into car and Truck using an ISA relationship

2. Common Attributes of Vehicle

   * Vehicle_ID
   * Registration_Number
   * Manufacturer
   * Model
   * Year
   * Purchase_Price

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3. Specialized Attributes

car :

* Number_of_Doors
* Fuel_Type
* Seating_capacity

Truck :

* Load_capacity
* Truck_Type
* Number_of_Axles

4. Primary key

* Vehicle_ID is the primary key of Vehicle.
* Vehicle_ID is inherited by both car and Truck.

5 ISA Specialization

Vehicle is the supertype and car and Truck are its subtypes.

6 Disjoint or overlapping

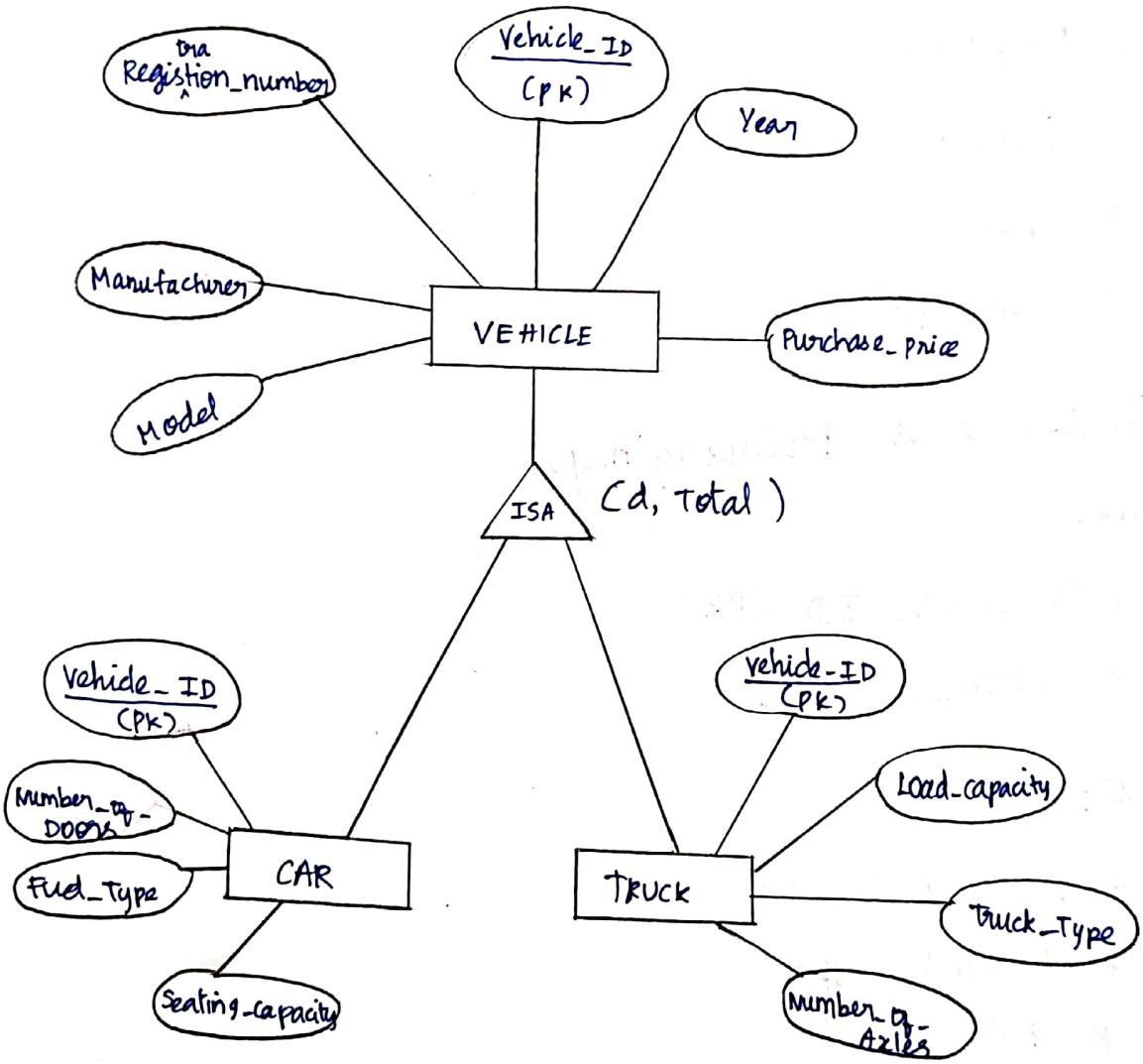
The specialization is Disjoint, because a vehicle can be either a car or a Truck, but not both.

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total or partial

The specialization is Total, because every Vehicle must belong to either Car or Truck.

8. EER Diagram



Here, d represents disjoint Specialization and Total means every Vehicle must belong to one subtype.

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3. Bank Management System

1. Major Entities

The major entities are :

* Customer
* Account
* Branch
* Employee
* Transaction
* Loan
* Loan_Payment

2&3 Attributes and Primary keys

Customer :

* Customer_ID (PK)
* Customer_Name

Account :

* Account_NO (PK)
* Account_Type
* Balance
* Opening_Date

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Branch :

* Branch_ID (PK)
* Branch_Name
* Location
* IFSC_code

Employee :

* Employee_ID (PK)
* Employee_Name
* Designation
* Salary

Transaction :

* Transaction_ID (PK)
* Transaction_Date
* Transaction_Type
* Amount

Loan

* Loan_ID (PK)
* Loan_Type
* Loan_Amount
* Interest_Rate
* Loan_Status

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Loan_Payment :

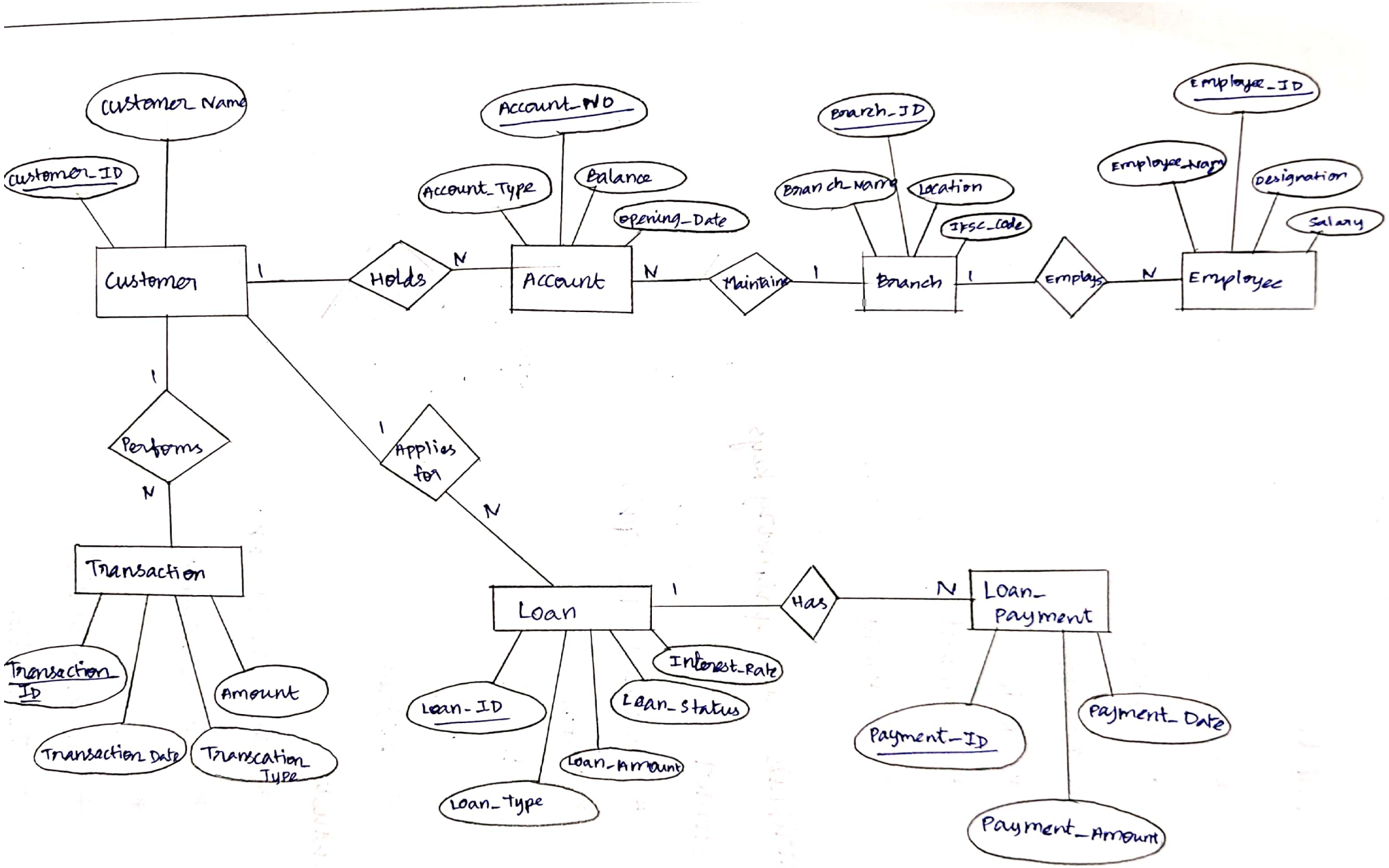
* Payment_ID (PK)
* Payment_Date
* Payment_Amount

4 & 5 Relationships and cardinalities :

Relationship	Cardianality
Customer_holds_~~Amount~~ Account	1:N
Branch_maintains_Account	1:N
Branch - employs - Employee	1:N
customer - performs - Transaction	1:N
customer - applies for - Loan	1:N
Loan - has - Loan_payment	1:N

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R Diagram



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E-commerce website :-

1 Major Entities

* customer
* Product
* category
* order
* Payment
* Delivery

2 Attributes and primary keys

Customer

* customer_ID (pk)
* Name
* Email
* Phone
* Address

Product

* Product_ID (pk)
* Product_Name
* Price
* Stock_quantity
* Description

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Category

* category_ID (PK)
* Category_Name

Order

* Order_ID (PK)
* Order_Date
* Order_status
* Total_Amount

Payment

* Payment_ID (PK)
* Payment_Date
* Payment_Mode
* Payment_status

Delivery

* Delivery_ID (PK)
* Delivery_Date
* Delivery_Status
* Delivery_Address

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Order - product Relationship

An order can contain many products and a product can appear in many orders. Therefore, the relationship is M:N.

The relationship has two attributes:

* Quantity
* Unit_price

An associative entity Order_Item can be used.

Order_Item:

* Order_ID (PK, FK)
* Product_ID (PK, FK)
* Quantity
* Unit_Price

4 & 5

Product specialization

Product can be specialized into:

* Electronics,
* clothing.
* Grocery.

The common Product attributes are inherited by all

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three subtypes.

Specialization constraint

A suitable choice is Disjoint and Total if every product belongs to exactly one category.

* Disjoint : A product belongs to only one category.

* Total : Every product must belong to a category

7 & 8 Relationships and cardinalities

* Customer — Places — order = 1:N
* Order — contains — product = M:N
* Order — has — Payment = 1:1
* Order — has — Delivery = 1:1
* Product — belongs to — category = N:1

9 EER Diagram:

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three subtypes.

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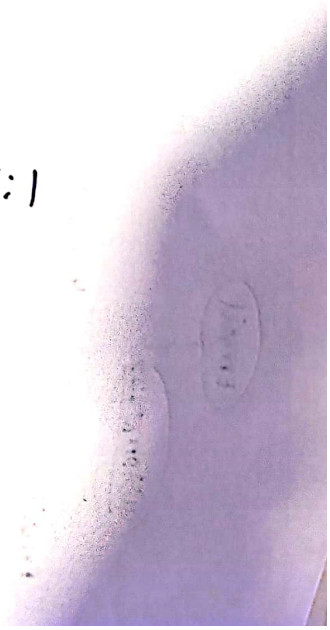
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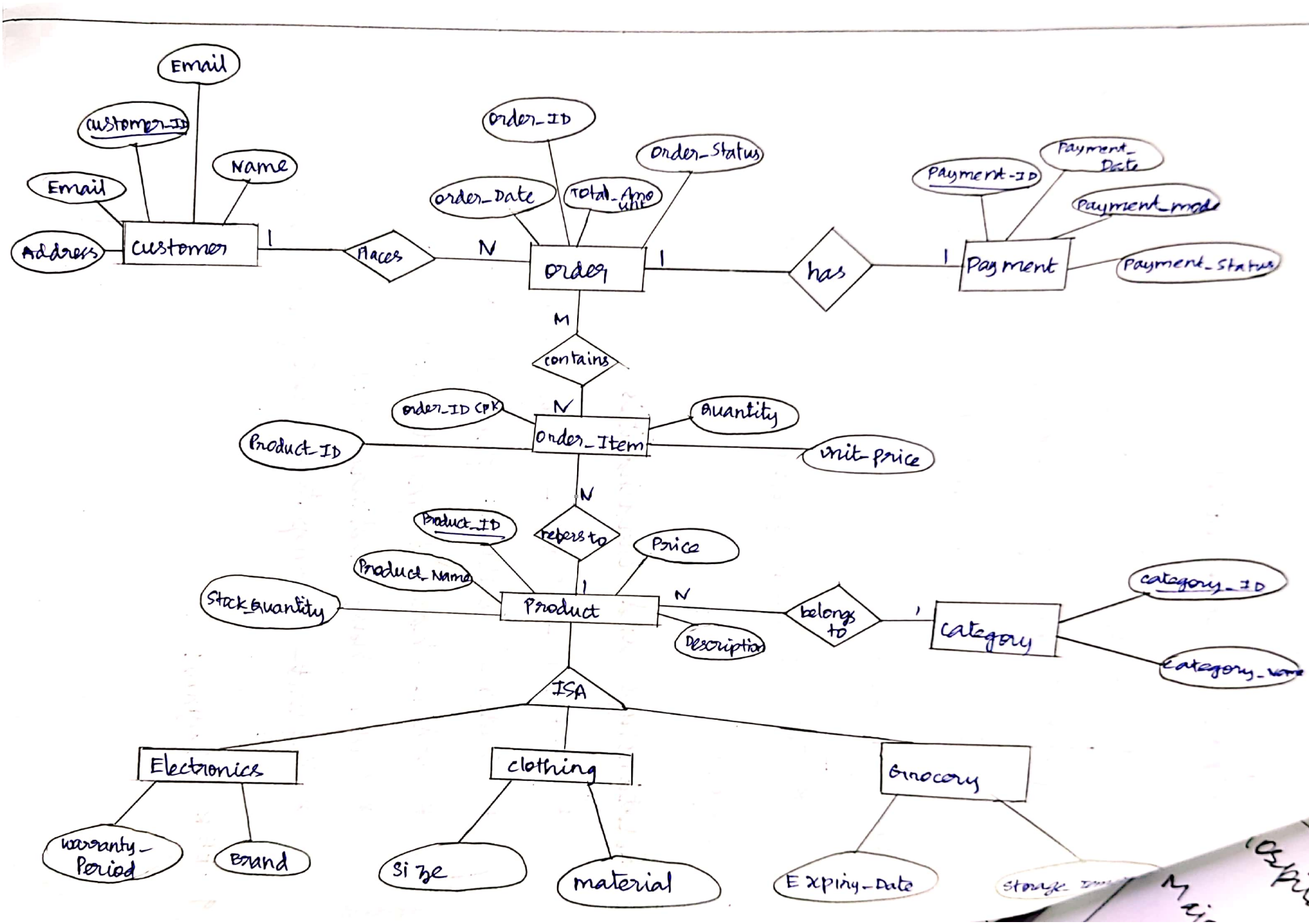
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9 EER Diagram:



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spital Management System

jor Entities

* Staff
* Patient
* Department
* Appointment
* Medical_Record
* Doctor
* Nurse
* Administrative_Staff

Staff specialization

The Staff Supertype is specialized into :

* Doctor
* Nurse
* Administrative_Staff

Inherited Attributes

All three subtypes inherit :

* Staff_ID (PK)
* Name
* Address
* Phone

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* Salary.

4 Specific Attributes

Doctor :

* Specilization
* License_Number

Nurse :

* Qualification
* shift

Administrative_Staff :

* Administrative_Specific attributes

5 Medical_Record as Weak Entity

Medical_Record is a weak entity because it depends on patient for its identification.

Attributes :

* Record_Date
* Diagnosis
* Treatment_Details.

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rtial Key

The Record_Date can be used as the partial key of Medical_Record.

Identifying Relationship

Patient has an identifying relationship with Medical_Record.

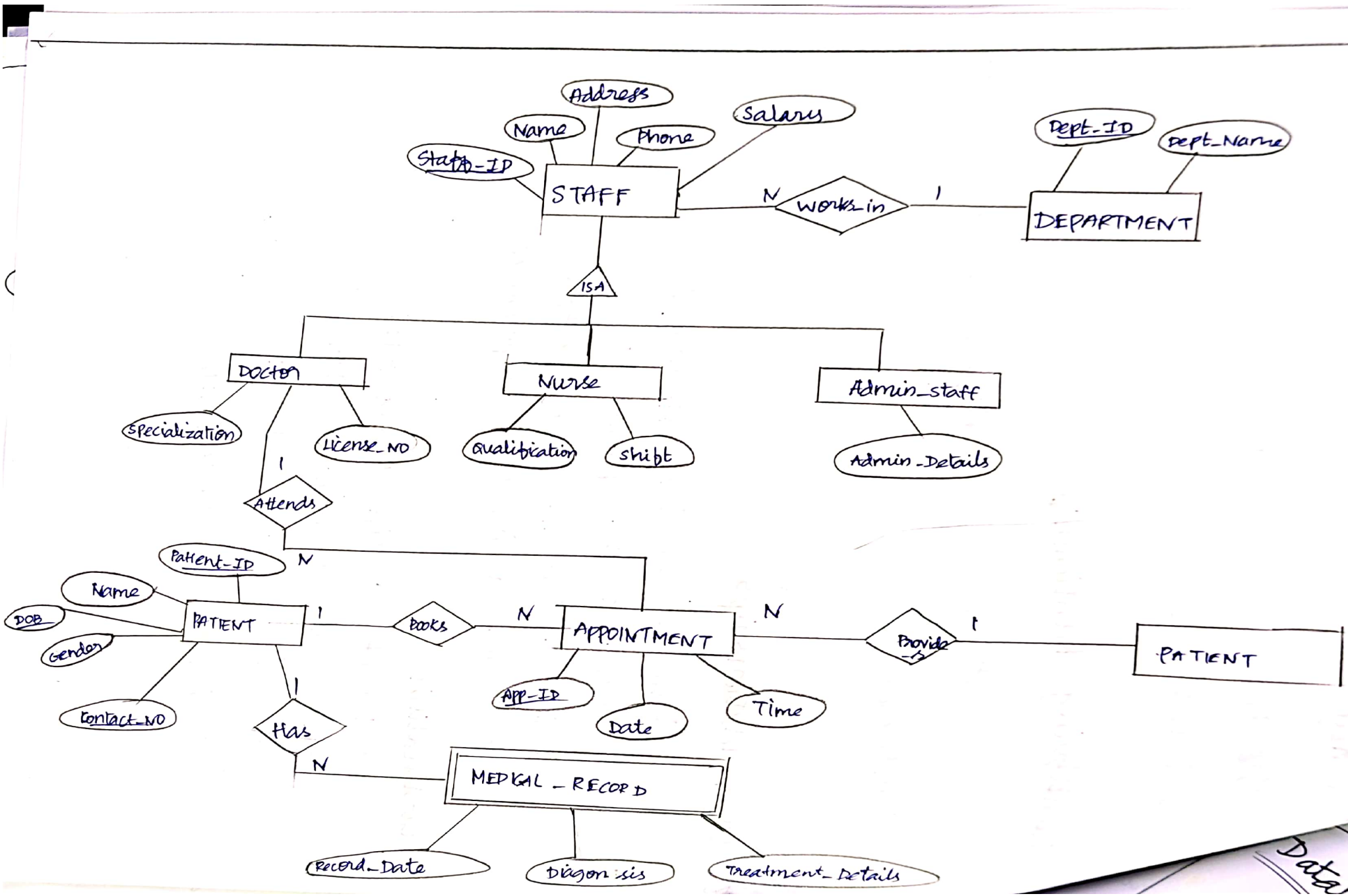
Other relationships

* Patient - books/has - Appointment
* Doctor - attends - Appointments
* Doctor - Provids - Treatment to Patient
* Doctor - assigned to - Department

Cardinalities

* Patient - Medical_Record = 1:N
* Patient - Appointment = 1:N
* Doctor - Appointment = 1:N
* Department - Doctor = 1:N

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Database Administrator (DBA)

Role of a DBA

A Database Administrator (DBA) is a person responsible for managing, maintaining, securing, and ensuring the reliable operation of an organization's database systems.

The DBA ensures that data is accurate, secure, consistent, available, and efficiently stored.

2. **Database Design and implementation :**

The DBA helps design the database structure according to organizational requirements. The DBA is involved in :

* Creating databases
* Designing tables and schemas
* Installing and configuring DBMs
* Implementing the database system

Monitoring database resources.

Troubleshooting and Disaster Recovery

The DBA identifies and resolves database Problems such as system failures, performance issues, and errors. Disaster recovery plans are used to restore database operations after major failures.

User and privilege Management

The DBA creates and manages user accounts and gives users appropriate privileges. Access is provided according to the user's role and requirements

Availability, Reliability and Security:

A DBA ensures datase availability by keeping the system accessible to authorised user. The DBA ensures reliability through backups, recovery, monitoring and maintenance. The DBA ensures Security through authentication, authorization, access control and Security policies.

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RICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

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